

ABHIJAY GANGARAPU

Independent Researcher

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EDUCATION

McNeil High School

Expected Graduation: May 2030

Two grades ahead in mathematics

Austin, TX

Incoming Freshman

RESEARCH INTERESTS

Number Theory, Algebra, p-adic Analysis, Algebraic Structures, Applied Mathematics, Computational Mathematics, Cryptography, Cybersecurity

PROJECTS

Period Growth in Piecewise Affine Systems over Residue Rings

2025–2026

Independent Research

- Proved theorems on super-linear period growth in piecewise affine systems over $\mathbb{Z}/p^k\mathbb{Z}$
- Demonstrated neural networks cannot learn sequences with Hensel-satisfied p-adic lifting properties
- Validated across 168 primes ($p = 2$ to $p = 997$) with mean growth ratio $26.94\times$
- Implemented in Python (NumPy, PyTorch, Numba JIT) with complete LaTeX documentation
- **GitHub:** github.com/CoderAwesomeAbhi/neural-cryptanalysis

Nexus Axiom: eBPF-Powered Linux Security Platform

2025–Present

Work in Progress — Cybersecurity Research

- Kernel-level runtime protection using eBPF LSM hooks and XDP for zero-overhead exploit prevention
- W^X memory protection that terminates processes attempting writable+executable memory allocations
- Real-time behavioral anomaly detection with ML-based threat prediction and MITRE ATT&CK correlation
- Network filtering with XDP, Prometheus metrics export, and integration with Slack/PagerDuty/Datadog
- Implemented in Rust (userspace) and C (eBPF kernel hooks) with comprehensive testing suite
- **GitHub:** github.com/CoderAwesomeAbhi/nexus-axiom

COMPETITIONS & AWARDS

Mathematics

- **AIME Qualifier** — American Invitational Mathematics Examination — 2026
- **AMC 10 Distinction** — Mathematical Association of America — 2026
- **MATHCOUNTS State Competitor** — Texas State Competition — 2026
- **Stanford Math Tournament (SMT) General:** Top 10% — 2026

Computer Science

- **USACO Silver Division** — USA Computing Olympiad — 2025–2026

Physics

- **USAPHO Qualifier** — USA Physics Olympiad — 2026

Science & Engineering

- **GARSEF 3rd Place Embedded Systems** — Engineering & General Sciences Division — 2026
- **GARSEF 4th Place** — Energy: Sustainable Materials and Design — 2024

TECHNICAL SKILLS

Programming: Python (advanced), Rust (intermediate), C/C++ (intermediate), JavaScript

ML/AI: PyTorch, TensorFlow, scikit-learn, Keras

Systems: eBPF, Linux kernel development, XDP, LSM hooks

Scientific Computing: NumPy, SciPy, Matplotlib, Numba (JIT compilation)

Mathematical Tools: LaTeX, Jupyter, Mathematica

Development: Git, GitHub, Linux, VS Code, Docker

RELEVANT COURSEWORK & SELF-STUDY

Completed: Geometry (two grades ahead in math), Computer Science (Python, C++, algorithms)

Self-Study: p-adic Number Theory (Gouvêa's *p-adic Numbers: An Introduction*), Abstract Algebra, Number Theory (Hensel's Lemma, arithmetic dynamics), Cryptography (pseudorandom generators, linear complexity), Linux Kernel Development, eBPF Programming

RESEARCH STATEMENT

I am passionate about applied mathematics, particularly number theory and algebraic structures, as well as systems security and kernel-level programming. I'm eager to learn from experienced researchers and contribute to ongoing research projects.

What I Seek: Research mentorship where I can learn rigorous mathematical techniques and contribute to your research program. I'm open to working on any project you suggest—I want to learn how professional mathematical research is conducted.

What I Offer: Strong work ethic, solid programming skills (Python, Rust, C), and genuine enthusiasm for mathematics and computer science. I'll do all coding, computations, and writing you need. I'm here to learn and contribute to your research goals.

Last Updated: May 14, 2026